b = allowance for bending (mm). The value of b shall be chosen so that the calculated stress in the bend, due to internal pressure only, does not exceed the allowable stress. Where such justification is not given, b shall be:
b = Dt0 / 2.5r (mm)
with:
r = mean radius of the bend (mm);
c = corrosion allowance (mm). If corrosion or erosion is expected the wall thickness of the piping shall be increased over that required by other
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thickness of the piping shall be increased over that required by other design regulations. This allowance shall be consistent with the expected life of the piping; and
= negative manufacturing tolerance for thickness (%), i.e. where is the manufacturing tolerance of -5%, || is equal to 5 and shall be entered into the formula as 1- (5/100)."
RESOLUTION MSC.551(108) (adopted on 23 May 2024)
AMENDMENTS TO THE INTERNATIONAL CODE OF SAFETY FOR SHIPS USING GASES OR OTHER LOW-FLASHPOINT FUELS (IGF CODE)

8 Bunkering
8.4 Regulations for manifold
11 Paragraph 8.4.1 is replaced by the following:
"8.4.1 The bunkering manifold shall be designed to withstand the external loads during bunkering. The connections at the bunkering station shall be arranged in order to achieve a dry-disconnect operation in one of the followings ways:
.1 a dry-disconnect / connect coupling in accordance with a standard at least equivalent to those acceptable to the Organization; or
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.1 a dry-disconnect / connect coupling in accordance with a standard at least equivalent to those acceptable to the Organization; or
.2 a manual connect coupler or hydraulic connect coupler, used to connect the bunker system to the receiving vessel bunkering manifold presentation flange; or
.3 a bolted flange to flange assembly.
12 The following new paragraphs are added after existing paragraph 8.4.1:
8.4.2 When intended to use either of the connections specified in paragraphs 8.4.1.2 and 8.4.1.3, these shall be combined with operating procedures that ensure a dry-disconnect is achieved. The arrangement shall be subject to special consideration informed by a bunkering arrangement risk assessment conducted at the design stage and considering dynamic loads at the bunkering manifold connection to a recognized standard acceptable to the Administration, the safe operation of the ship and other hazards that may be r elevant to the ship during bunkering operation.
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ship and other hazards that may be r elevant to the ship during bunkering operation. The fuel handling manual required by 18.2.3 shall include documentation that the bunkering arrangement risk assessment was conducted, and that special consideration was granted under this requirement.
8.4.3 An emergency release coupler (ERC) / Emergency Release System (ERS) or equivalent means shall be provided, unless installed on the bunkering supply side of the bunkering line, and the said means shall be in accordance with a standard equivalent to those acceptable to the Organization; it shal l enable a quick physical disconnection dry break-away of the bunker system in an emergency event."
9 Fuel supply to consumers
9.3 Regulations on redundancy of fuel supply
13 Paragraph 9.3.1 is replaced by the following:
"9.3.1 For ships constructed on or after 1 January 2026, for single fuel installations the fuel supply system shall be arranged with redundancy and segregation, so that a
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the fuel supply system shall be arranged with redundancy and segregation, so that a leakage in one system, or failure of one of the fuel supply essential auxiliaries, does not lead to an unacceptable loss of power. In the event of a leakage or failure, and in accordance with SOLAS regulation II -1/26.3, the Administration, having regard to overall safety considerations, may accept a partial reduction in propulsion capabili ty from normal operation."
RESOLUTION MSC.551(108) (adopted on 23 May 2024)
AMENDMENTS TO THE INTERNATIONAL CODE OF SAFETY
FOR SHIPS USING GASES OR OTHER LOW-FLASHPOINT FUELS (IGF CODE)

9.4 Regulations on safety functions of gas supply system
14 Paragraph 9.4.7 is replaced by the following:
"9.4.7 For ships constructed on or after 1 January 2026, in cases where the master gas fuel valve is automatically shut down when the safety system as required in 15.2.2 is activated, the complete gas supply pipe between this master gas fuel valve and the
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is activated, the complete gas supply pipe between this master gas fuel valve and the double block and bleed valves and between the double block and bleed valves and the consumer shall be automatically vented."
15 Paragraph 9.4.8 is replaced by the following:
"9.4.8 For ships constructed on or after 1 January 2026, there shall be one manually operated shutdown valve in the gas supply line to each gas consumer upstream of the double block and bleed valves to assure safe isolation during maintenance on the gas consumer."
9.6 Regulations for fuel supply to consumers in gas-safe machinery spaces
16 Paragraph 9.6.1.1 is replaced by the following:
"9.6.1 Gas fuel piping in gas-safe machinery spaces shall be completely enclosed by a double pipe or duct fulfilling one of the following conditions:
.1 the gas fuel piping shall be a double wall piping system with the gas

fuel contained in the inner pipe. The space between the concentric
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.1 the gas fuel piping shall be a double wall piping system with the gas
fuel contained in the inner pipe. The space between the concentric
pipes shall be pressurized with inert gas at a pressure greater than
the gas fuel pressure. Suitable alarms shall be provided to indicate
a loss of inert gas pressure between the pipes; or"
9.8 Regulations for the design of ventilated duct, outer pipe against inner pipe gas
leakage
17 Paragraph 9.8.1 is replaced by the following:
"9.8.1 For ships constructed on or after 1 January 2026, the design pressure of the
outer pipe or duct of fuel systems shall not be less than the maximum working
pressure of the inner pipe. Alternatively, the design pressure of the outer pipe or duct
may be calculated in accordance with 9.8.2."
18 The chapeau of paragraph 9.8.2 is replaced by the following:
"9.8.2 For ships constructed on or after 1 January 2026, alternatively to 9.8.1, the
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"9.8.2 For ships constructed on or after 1 January 2026, alternatively to 9.8.1, the
design pressure of the outer pipe or duct shall be taken as the higher of the following:"
19 Paragraph 9.8.4 is replaced by the following:
"9.8.4 For ships constructed on or after 1 January 2026, the duct shall be pressure-
tested to show that it can withstand the expected maximum pressure at fuel pipe
rupture."
11 Fire safety
11.3 Regulations for fire protection
20 Paragraph 11.3.1 is replaced by the following:
"11.3.1 For ships constructed on or after 1 January 2026, fuel preparation rooms
shall, for the purpose of the application of SOLAS regulation II-2/9, be regarded as a
machinery space of category A."
RESOLUTION MSC.551(108) (adopted on 23 May 2024)
AMENDMENTS TO THE INTERNATIONAL CODE OF SAFETY
FOR SHIPS USING GASES OR OTHER LOW-FLASHPOINT FUELS (IGF CODE)
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11.6 Regulations for dry chemical powder fire-extinguishing system
21 Paragraph 11.6.2 is replaced by the following:
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- 8 -
11.6 Regulations for dry chemical powder fire-extinguishing system
21 Paragraph 11.6.2 is replaced by the following:
"11.6.2 In addition to any other portable fire extinguishers that may be required
elsewhere in IMO instruments, one portable dry powder extinguisher of at least 5 kg
capacity shall be located near the bunkering station and in the fuel preparation room.

For ships constructed before 1 January 2026, the portable dry powder extinguisher shall be provided in the fuel preparation room not later than the first survey on or after 1 January 2026."

12 Explosion prevention

12.5 Hazardous area zones

22 Paragraph 12.5.1 is replaced by the following:

"12.5.1 Hazardous area zone 0

For ships constructed on or after 1 January 2026, this zone includes, but is not limited to, the interiors of fuel tanks, any pipework for pressure relief or other venting systems for fuel tanks, pipes and equipment containing fuel, and interbarrier spaces as defined

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for fuel tanks, pipes and equipment containing fuel, and interbarrier spaces as defined by paragraph 2.2.15.2."

12.5.2 Hazardous area zone 1

23 Paragraph 12.5.2.1 is replaced by the following:

".1 for ships constructed on or after 1 January 2026, tank connection spaces and fuel storage hold spaces;"

15 Control, monitoring and safety systems

15.4 Regulations for bunkering and liquefied gas fuel tank monitoring

15.4.1 Level indicators for liquefied gas fuel tanks

24 Paragraph 15.4.1.3 is replaced by the following:

".3 For ships constructed on or after 1 January 2026, liquefied gas fuel tank liquid level gauges may be of the following types:

.1 indirect devices which determine the amount of fuel by means such as weighing or in-line flow metering;

.2 closed devices which do not penetrate the liquefied gas fuel tank, such as devices using radioisotopes or ultrasonic devices; or

.3 closed devices which penetrate the liquefied gas fuel tank but which

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such as devices using radioisotopes or ultrasonic devices; or

.3 closed devices which penetrate the liquefied gas fuel tank but which form part of a closed system and keep the gas fuel from being released. Such devices shall be considered as tank connections. If the closed gauging device is not mounted directly onto th e tank, it shall be provided with a shut off valve located as close as possible